

Statement of Work (SOW)

Project Title: Student Performance Visualization Tool

Date: 09/02/2026

Prepared By: Mahindra University

1. Introduction:

This Statement of Work (SOW) outlines the objectives, scope, deliverables, timeline, and responsibilities for the **Student Performance Visualization Tool**, to be developed as part of the Software Engineering course at Mahindra University.

The purpose of this project is to design and develop a software application that analyzes student academic data and presents performance insights through graphical visualizations. The tool will help students and instructors easily understand academic progress and identify strengths and areas for improvement.

2. Scope of Work:

The Students Performance Visualization Tool is a web-based or application-based system designed to manage and visualize students' academic data. It will allow users to input marks, attendance, and semester information, and generate visual reports for analysis.

Involved.

Objectives:

- To collect and manage student academic performance data.
- To calculate performance metrics such as total marks, average, percentage, and grades.
- To visualize student performance using graphical representations like bar charts, pie charts, and line graphs.
- To analyze individual and class performance for better understanding of strengths and weaknesses.
- To develop the system using software engineering principles such as modularity, usability, and maintainability.

Key Activities:

- Requirement analysis and system planning
- Designing system architecture and database schema
- Front-end interface development
- Back-end logic implementation
- Data visualization integration
- Testing and debugging
- Final deployment and documentation

3. Deliverables

Deliverable 1: Software Requirements Specification (SRS) and Project Planning – 10 Feb 2026

Deliverable 2: System Design and Architecture Document – 20 Feb 2026

Deliverable 3: Data Input and Processing Module – 9 Mar 2026

Deliverable 4: Visualization Module (implementation of graphs for performance analysis) – 21 Mar 2026

Deliverable 5: Student Search and Class Analysis Features (class average, top performers, and performance comparison) – 6 Apr 2026

Deliverable 6: User Interface Integration – 17 Apr 2026

Deliverable 7: Testing and Validation Report – 30 Apr 2026

Deliverable 8: Final Working Application with Sample Dataset and User Guide/README – 5 May 2026

Deliverable 9: Final Project Report and Presentation – 12 May 2026

4. Timeline and Milestones

Provide a high-level timeline with key milestones.

Milestone	Description	Due Date
Project Kickoff	Initial project planning, topic finalization, and requirement discussion	10 Feb 2026

Milestone	Description	Due Date
Phase 1 Completion	Requirements analysis, system design, and data structure preparation	20 Feb 2026
Phase 2 Completion	Implementation of data input and performance calculation modules	9 Mar 2026
Phase 3 Completion	Development of visualization features (bar chart, pie chart, line graph)	21 Mar 2026
Phase 4 Completion	Implementation of student search, class average, and comparison features	6 Apr 2026
Phase 5 Completion	User interface integration and full system functionality	30 Apr 2026
Phase 6 Completion	Data security features will be basic and not enterprise-grade.	5 May 2026
Final Delivery	Final submission of working application, source code, dataset, and project report	12 May 2026

5. Roles and Responsibilities

Outline the key roles and their responsibilities within the project.

Team Members:

- M.LEELA KARTHIKEYA - Project manager
- E.PRIYAMVADA - Requirement Analyst
- S.SRI SAMPATH - System Designer
- NAVVYAH - Front-End Developer
- M.SATWIK REDDY - Back-End Developer
- A.NEAL TEJA - Database Administrator

- K.SATYA VARDHAN - Data Visualization & Analytics
 - A.SATHVIK REDDY - Testing & Quality Assurance
 - **Client Contact:** Software Engineering course, Mahindra University
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7. Assumptions and Constraints

Assumptions:

- Student academic data will be available in structured format (CSV, Excel, or manual input).
- Users (students/faculty) will provide accurate and complete data for analysis.
- Development tools such as Python, web frameworks, and visualization libraries will be accessible.
- The system will be used within an academic environment for demonstration and learning purposes.
- The application will run on standard systems with basic internet support.
- Users will have basic knowledge of operating web-based applications.

Constraints:

- Development time is limited due to academic semester deadlines.
- The system is designed for class-level datasets and may not scale for institution-wide deployment.
- Real-time database hosting and cloud deployment may not be implemented due to infrastructure limitations.
- Advanced predictive analytics or AI-based forecasting features are optional and may be excluded based on time constraints.
- The application is a prototype and not intended for commercial or production-level use.
- Data security features will be basic and not enterprise-grade.

8. Approval Signatures

Client: Mahindra University

Title: Student Performance Visualization Tool

Date: 09/02/2026

Service Provider: Team No - 39

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